Calculation of I^2 in the Bayesian Framework

This supplementary file describes how I^2 was calculated for the Bayesian meta-analysis.

Higgins & Thompson (2002) define I^2 as the proportion of total variance in observed effect sizes that is attributable to between-study heterogeneity:

$$I^2 = \frac{\tau^2}{\tau^2 + v_t}$$

where τ^2 is the between-study variance and v_t is a measure of typical within-study variance across k included studies.

v_t is a weighted summary of all within-study sampling variances σ_k^2, computed by `metafor` (Viechtbauer 2010) as:

$$v_t = \frac{(k-1)\sum_k w_k}{\left(\sum_k w_k\right)^2 - \sum_k w_k^2}$$

where $w_k = 1/\sigma_k^2$ are the inverse-variance weights.

The Bayesian meta-analysis produces a full posterior distribution of τ and doesn't yield a single point estimate of v_t. To ensure that I^2 is computed on the same scale and with the same denominator in both frameworks, v_t is held fixed at the value obtained from `metafor` ($v_t = 0.59$). This is a deliberate methodological choice: v_t depends on the within-study sampling variances σ_k^2, which are treated as known constants in both models (see model specification in main article, 2.4 Methods). Reusing v_t ensures that any difference between the two I^2 estimates reflects only the difference in the estimated τ^2, not a difference in the denominator.

Let $\tau^{(s)}$, for $s = 1, \ldots, S$, denote the posterior draws of the between-study standard deviation obtained from the fitted Bayesian model. The between-study variance for each draw is:

$$\tau^{2(s)} = \left(\tau^{(s)}\right)^2$$

Substituting into the standard formula with the fixed v_t, I^2 is computed for each posterior draw as:

$$I^{2(s)} = \frac{\tau^{2(s)}}{\tau^{2(s)} + v_t}$$

This yields a posterior distribution $\{I^{2(s)}\}_{s=1}^{S}$ for I^2. The posterior mean is:

$$\mathbb{E}[I^2 \mid \text{data}] \approx \frac{1}{S} \sum_{s=1}^{S} I^{2(s)}$$

and a 95% credible interval is obtained from the 2.5th and 97.5th quantiles of this same set of draws:

$$95\% \text{ CrI} = \left[Q_{0.025}\left(\{I^{2(s)}\}\right),\ Q_{0.975}\left(\{I^{2(s)}\}\right)\right]$$

References

Higgins, J. P., & Thompson, S. G. Quantifying heterogeneity in a meta-analysis. Statistics in medicine, 2002;21:1539–1558. https://doi.org/10.1002/sim.1186.

Wolfgang Viechtbauer. Conducting meta-analyses in with the metafor package. Journal of Statistical Software 2010;36:1–48. https://doi.org/10.18637/jss.v036.i03.